

Operational Risk & Trust Integrity Standard (ORTIS)

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Operational Risk & Trust Integrity Standard -

(ORTIS)

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Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Operational Reality Standards™

Operational Layer: Autonomous Systems Governance Layer

Governed Space: Operational Risk & Trust Integrity

Category: Governance & Enforcement

Subcategory: Operational Risk & Trust Governance Architecture

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Compatibility: OOF Methodology OS · Runtime Integrity Standard (RIS) · Operational Authority Integrity Standard (OAIS) · Operational Evidence & Auditability Standard (OEAS) · Operational Escalation Integrity Standard (OESIS) · Operational Constraint Integrity Standard (OCNS) · Operational Decision Integrity Standard (ODIS) · INTEGROS® — Integrity Standard · Multi-Layer Truth Validation Framework (MTVF) · Ethical Virtual Integrity Protocol (EVIP)

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition System

Canonical Definition

Operational Risk & Trust Integrity Standard (ORTIS) defines the structural conditions under which operational trust, runtime risk, permission eligibility, sensitivity classification, trust degradation, trust recovery, and consequence-bearing risk states remain materially stable, traceable, governable, and operationally valid across autonomous and semiautonomous operational environments. ORTIS governs the relationship between what a system is trusted to do, what risk it carries, and whether it remains eligible to execute actions under defined operational conditions.

Operational validity increasingly depends not only on execution capability, authority, or evidence, but also on whether risk and trust remain materially aligned during runtime operation.

A. Standard Abstract

- Future operational systems increasingly operate through:
- autonomous AI agents
- multi-agent environments
- tool-using systems
- adaptive runtime systems
- delegated execution environments
- machine-to-machine coordination
- autonomous decision infrastructures
- consequence-bearing operational workflows
- Yet most systems still treat trust and risk as:
- static permissions
- user roles
- access levels
- risk scores
- security labels
- approval states
- This creates a major governance problem.
- A system may preserve runtime execution while operational trust and runtime risk progressively diverge.
- An agent may still have permission to act while its trust state has degraded.

A tool may still be accessible while the risk level has changed.

A decision may still execute while the operational risk has exceeded governance-valid conditions. ORTIS exists to govern that condition.

B. Core Principle

Operational trust is not valid merely because access is granted.

Operational risk is not valid merely because execution is possible. Risk and trust become operationally valid only when trust level, risk state, sensitivity condition, permission eligibility, and consequence-bearing execution remain materially aligned during runtime operation.

C. Scope

This standard may apply to: autonomous AI systems

AI agents

- multi-agent infrastructures
- enterprise AI environments
- tool-using agent systems
- runtime execution platforms
- delegated authority systems
- autonomous robotics
- financial automation systems
- healthcare decision-support systems
- critical infrastructure environments
- machine-governed operational systems

ORTIS applies wherever operational validity depends on the alignment of

risk, trust, permission, and consequencebearing execution.

D. Why This Standard Exists

Autonomous systems increasingly perform actions under changing operational conditions. Trust may change. Risk may change. Context may change. Sensitivity may change.

Authority may change.

Execution eligibility may change. Traditional access-control models often answer: Is this action allowed? ORTIS asks a deeper operational question: Is this action still governance-valid under the current risk and trust state? This distinction becomes critical when autonomous systems execute actions dynamically across tools, environments, users, permissions, and operational contexts.

A system may remain technically authorized while no longer being trust-valid or risk-valid.

ORTIS exists because operational trust and runtime risk become governable spaces in autonomous systems.

E. Operational Risk & Trust Logic

Operational risk and trust integrity exists only when the following remain materially preservable:

1. Trust State Integrity

The operational trust level remains materially aligned with current system behavior, authority, context, and evidence.

2. Runtime Risk Integrity

Runtime risk remains classified, bounded, traceable, and operationally governable.

3. Permission Eligibility Integrity

Execution eligibility remains aligned with trust level, risk level, sensitivity, authority, and operational context.

4. Trust Degradation Integrity

Trust reduction, trust loss, and trust restriction remain detectable, traceable, and governable.

5. Consequence-Bearing Risk Integrity

Risk states affecting real operational outcomes remain materially accountable, auditable, and governance-valid.

F. Operational Architecture Space

- ORTIS defines the operational architecture space for:
- operational trust governance
- runtime risk governance
- permission eligibility governance
- sensitivity-based execution governance
- trust degradation governance
- trust recovery governance
- autonomous risk classification
- risk escalation
- consequence-bearing risk governance
- trust-risk alignment
- This space exists because future autonomous systems increasingly require governance not only of what they can do, but
- whether they should still be trusted to do it under current operational risk conditions.

G. Difference Between Access Control and Risk

& Trust Integrity Access control and operational risk-trust integrity are related but structurally distinct. Access control governs: whether access is granted whether a role has permission whether a credential is valid whether a system may technically perform an action

Operational risk and trust integrity governs:

- whether the system remains trust-valid
- whether risk remains acceptable
- whether permission remains contextually legitimate
- whether trust has degraded
- whether execution remains consequence-valid

A system may have access while no longer being trust-valid.

A system may be authorized while no longer being risk-valid.

ORTIS therefore governs trust-risk legitimacy itself.

H. Runtime Position

ORTIS operates within the Autonomous Systems Governance Layer. It sits alongside runtime execution, authority, escalation, constraint, and evidence governance. ORTIS governs whether operational permission remains valid when trust and risk states change during runtime operation. This distinction becomes critical in:

AI agent tool use

- autonomous execution
- delegated authority environments
- sensitive operational workflows

- high-risk autonomous systems
- multi-agent infrastructures
- consequence-bearing decision environments

I. Trust-Risk Drift Rule

Trust-risk drift occurs when operational trust, runtime risk, permission eligibility, or sensitivity conditions progressively diverge while systems continue assuming execution remains valid. This may include: trust degradation without permission restriction risk escalation without execution limitation outdated trust classification static access under dynamic risk sensitivity mismatch permission eligibility collapse consequence-bearing risk misalignment A system may continue operating while trust-risk validity underneath has already degraded materially. ORTIS exists to expose and govern that condition.

J. Validity Logic

A system is valid under ORTIS when:

- operational trust remains materially aligned with current conditions
- runtime risk remains classified and governable
- permission eligibility remains contextually valid
- trust degradation is detectable and actionable
- risk escalation remains traceable
- consequence-bearing risk remains accountable
- execution remains aligned with trust-risk conditions

A system becomes invalid under ORTIS when:

- trust state becomes outdated or materially inaccurate
- risk state becomes unclassified or uncontrolled
- permission remains active despite trust degradation
- execution continues despite unacceptable risk
- sensitivity conditions are ignored
- consequence-bearing risk cannot be audited
- trust-risk alignment cannot be demonstrated

K. Relationship to Other OOF Standards

ORTIS operates naturally with:

Runtime Integrity Standard (RIS)

Operational Authority Integrity Standard (OAIS)

Operational Evidence & Auditability Standard (OEAS)

Operational Escalation Integrity Standard (OESIS)

Operational Constraint Integrity Standard (OCNS)

Operational Decision Integrity Standard (ODIS)

INTEGROS® — Integrity Standard Multi-Layer Truth Validation Framework (MTVF) Ethical Virtual Integrity Protocol (EVIP) ORTIS does not replace these standards. It governs the operational risk and trust layer through which execution eligibility remains materially aligned with risk, trust, sensitivity, authority, and consequence-bearing conditions.

L. Foundational Principle

If trust and risk cannot remain materially aligned during runtime operation, systems may preserve execution while governance-valid permission, legitimacy, and consequence-bearing operational integrity progressively destabilize.

Canonical Closing Statement

Operational Risk & Trust Integrity Standard (ORTIS) defines the structural conditions under which operational trust, runtime risk, permission eligibility, sensitivity classification, trust degradation, trust recovery, and consequence-bearing risk states remain materially stable, traceable, governable, and operationally valid across autonomous and semiautonomous operational environments. Operational systems are not valid merely because access exists or execution is technically possible. Operational validity increasingly depends on whether trust, risk, permission, and consequence-bearing execution remain materially aligned during runtime operation. Nahlad

Operational Risk & Trust Integrity Standard -

(ORTIS)

Architecture Family: Operational Reality Standards™

Operational Layer: Autonomous Systems Governance Layer

Category: Governance & Enforcement

Subcategory: Operational Risk & Trust Governance Architecture

Defines the conditions under which operational trust, runtime risk, permission eligibility, sensitivity classification, trust degradation, and consequence-bearing risk states remain traceable, governable, and operationally valid across autonomous and semi-autonomous operational environments.

ORTIS addresses trust degradation, risk escalation, permission eligibility, sensitivity mismatch, trust-risk drift, and execution legitimacy under changing runtime conditions.

Relevant for AI agents, multi-agent systems, autonomous execution environments, enterprise AI infrastructures, delegated

authority systems, robotics ecosystems, and consequence-bearing autonomous operations.

→ View Standard

About standards

Module Architecture

→ ATRIM — Agent Trust & Risk Integrity Module

→ REM — Risk Escalation Module

→ TDM — Trust Degradation Module

→ TRM — Trust Recovery Module

→ ARCM — Autonomous Risk Classification Module

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Canonical Source

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